DARSHAN S

AI/ML Engineer
Bangalore, Karnataka, India

+91-9108232080
darshans31805@gmail.com
GitHub Portfolio LinkedIn

Professional Summary

AI & ML undergraduate with experience building computer vision systems for geospatial property analysis and document-grounded RAG applications. Skilled in Python, FastAPI, React, semantic caching, intelligent LLM routing, model evaluation, and full-stack development. Focused on developing practical AI systems with measurable cost optimization, reliable retrieval, and observability.

Education

M S Ramaiah University of Applied Sciences — In Progress
B.Tech, Artificial Intelligence & Machine Learning — CGPA: 8.0

Christ Junior College — 2023
Pre-University Course, KSEAB, Karnataka — 83.6%

Cambridge Public School — 2021
10th Boards, CISCE — 92.8%

Experience

MPRO9 — Jul. 2026 – Oct. 2026
AI Intern – Project Development & Implementation — *Mysore, India*

- Built GeoVision AI for property identification and analysis using geospatial and visual data.
- Developed YOLO-based object detection workflows for analyzing 360-degree imagery.
- Integrated visual data with GPS information using Insta360 X4 imagery and u-blox GPS modules.
- Contributed to data preprocessing, model development, testing, and optimization.

Personal Projects

DECIPHER – Meeting-to-Execution Agent
Next.js, React, TypeScript, Tailwind CSS, Supabase, PostgreSQL, Zod, Google Gemini API, OpenRouter, Trello API, Vercel

- Built an AI-powered meeting intelligence platform that transforms meeting transcripts into structured decisions, commitments, risks and dependencies, each linked to its source transcript line.
- Implemented human-in-the-loop approval workflows with transcript validation before saving information or sending it to Trello.
- Developed cross-meeting decision intelligence with a Decision Timeline, Commitment Tracker, Impact Analysis and AI Command Center, integrated with Trello for approved commitments.

Cost-Aware RAG – Semantic Caching & Intelligent Model Routing
Python, FastAPI, React, TypeScript, SQLite, Groq, Sentence Transformers

- Built a document-grounded RAG pipeline with PDF/TXT ingestion, page-aware chunking, vector retrieval, source citations, and insufficient-evidence handling, exposed through 13 REST endpoints.
- Implemented evidence-aware semantic caching and complexity-based LLM routing, reducing estimated inference cost by 59.2% and model calls from 38 to 31 in a 38-request offline evaluation with zero incorrect cache hits.
- Created isolated temporary visitor workspaces using server-issued tokens and in-memory SQLite, with session-scoped documents, retrieval, caching, and telemetry; validated with 145 passing automated tests.

Smart Traffic Vehicle Counter
Python, YOLO11, ByteTrack, OpenCV, Streamlit

- Developed a vehicle detection, tracking, classification, and counting system for 4 vehicle classes with direction and speed estimation, organized into 7 modules with 21 unit tests.
- Combined YOLO11, ByteTrack, trajectory analysis, and line-crossing logic in a Streamlit analytics dashboard; benchmarked ByteTrack against BoT-SORT on a traffic video (43 vs. 75 track IDs, 4 vs. 26 short fragmented tracks).

AI Health Tracker
React.js, Node.js, Express.js, MongoDB, Google Gemini API, JWT

- Built a full-stack healthcare platform with 4 tracking modules and Gemini-powered personalized recommendations.
- Implemented JWT authentication, 19 REST API endpoints, and MongoDB persistence across 5 data models, with a 9-page responsive React dashboard, Docker setup, and GitHub Actions CI.

Technical Skills

Languages: Python, JavaScript, Java, C
Frontend: React.js, TypeScript, HTML5, CSS3, Tailwind CSS
Backend: FastAPI, Node.js, Express.js, REST APIs
AI/ML: RAG, Semantic Caching, LLM Routing, Sentence Transformers, TensorFlow, Scikit-learn, YOLO, Computer Vision, NLP, Deep Learning, LangChain, LangGraph, Google Gemini API
Databases: SQLite, MongoDB, SQL
Tools: Git, GitHub, Postman, VS Code, Jupyter Notebook, Pytest, OpenCV, Streamlit
Hardware: Insta360 X4, u-blox GPS
Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Operating Systems, Computer Networks

Certifications

- Oracle Certified: AI Vector Search Certified Professional
- AI Fluency Framework and Foundations – Anthropic
- AI Literacy – IBM
- AI Fundamentals with IBM SkillBuild – Cisco
- Data Analytics Essentials – Cisco
- Prompt Design in Vertex AI – Google
- Generative AI Foundations – upGrad in collaboration with Microsoft

Achievements

- Participant – Agentic AI Hackathon | Product Space × Wispr Flow (2026): developed and deployed DECIPHER, an AI meeting-intelligence platform, as part of the hackathon
- Participant – Smart India Hackathon
- Participant – Tata Elxsi Hackathon